

LEAD AGENT

Project Knowledge Document

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Executive Overview

Lead Agent is a production-ready, AI-powered inbound lead qualification and research system deployed on Vercel. It automates the entire lead screening pipeline—from initial form submission through verification, research, qualification, and approval workflows—while maintaining human oversight through Slack integration. The system processes 10+ leads per minute with minimal latency, enabling sales teams to focus on qualified prospects.

This document serves as the complete technical reference for understanding, deploying, and customizing the Lead Agent system.-

Key Highlights

- Real-time lead verification via 6 parallel verification modules
- AI-powered deep research using Vercel AI SDK with multi-model support
- Durable workflow execution via Workflow DevKit (survives infrastructure failures)
- Slack-based human-in-the-loop approval system
- Web research capability via Exa.ai integration
- Production deployment on Vercel with auto-scaling

1. Project Overview

Purpose

Lead Agent is an inbound lead qualification and research system that:

1. Captures leads via web form
2. Performs real background verification and research
3. Qualifies leads based on company research and decision-maker validation
4. Generates personalized emails through human-in-the-loop approval
5. Integrates with Slack for team collaboration

Business Value

- Automated qualification reduces manual lead screening time
- Real research validates lead quality before sales engagement
- Human-in-the-loop prevents false positives
- Scalable - processes multiple leads in parallel via Workflow DevKit

Target Users

Sales and business development teams using Slack for qualifications and approvals.

2. Technology Stack

Core Framework

Technology	Version	Purpose
Next.js	16.0.10	Full-stack framework, API routes, Server components
React	19.2.1	Frontend UI components
TypeScript	5.x	Type safety across codebase

AI & Workflow

Technology	Version	Purpose
Vercel AI SDK	5.0.68	LLM integration, structured generation (generateObject)
Workflow DevKit	4.0.1-beta.24	Durable background task execution
Vercel AI Gateway	N/A	Multi-model routing, cost optimization

Integrations

Technology	Version	Purpose
Slack Bolt	4.6.0	Slack bot framework
Vercel Slack Bolt Adapter	1.0.2	Serverless Slack integration
Exa.ai SDK	2.0.11	Web search & content extraction for research

Infrastructure

Technology	Version	Purpose
Vercel KV	1.0.1	Redis-compatible key-value storage
Vercel Deployment	N/A	Hosting, Functions, Edge middleware

UI Components

Technology	Version	Purpose
Radix UI	Latest	Headless UI components (form, label, separator)
Tailwind CSS	4	Utility-first CSS
React Hook Form	7.64.0	Form state management
Sonner	2.0.7	Toast notifications

3. Architecture Overview

Data Flow

-
- User Form Submission
- ↓
- POST /api/submit (form data validation)
- ↓
- Create Workflow Instance (Workflow DevKit)
- ↓
- Background Steps Execute in Parallel
 - └─ stepResearch (Background Verification)
 - └─ performBackgroundVerification()
 - └─ deepResearch() [AI Agent]
 - └─ compileResearchReport()
 - └─ stepQualify (Lead Qualification)
 - └─ generateObject() [LLM]
 - └─ stepEmail (Email Generation)
 - └─ writeEmail() [LLM]
 - └─ humanFeedback() [Slack]
 - └─ User approves/denies
 - └─ Webhook triggers emailResponse()
 - └─ Send email if approved

Component Hierarchy

- - Next.js Application
 - | — Frontend Layer
 - | | — app/page.tsx (Homepage, Lead Form)
 - | | — components/lead-form.tsx (Form UI)
 - | | — components/ui/ (Radix UI components)
 - | — API Layer
 - | | — app/api/submit/route.ts (Lead ingestion)
 - | | — app/api/slack/route.ts (Slack webhook receiver)
 - | — Workflow Layer
 - | | — workflows/inbound/steps.ts (Step functions)
 - | | — stepResearch (Background verification)
 - | | — stepQualify (Lead qualification)
 - | | — stepEmail (Email generation)
 - | | — humanFeedback (Slack approval)
 - | — Service Layer
 - | | — lib/services.ts (AI service calls)
 - | | — lib/background-verification.ts (Verification logic)
 - | | — lib/exa.ts (Web research)
 - | | — lib/slack.ts (Slack messaging)
 - | — Utilities
 - | | — lib/types.ts (Type definitions)
 - | | — lib/utils.ts (Helper functions)
 - | | — lib/qualification-rules.ts (Qualification logic)

4. Key Components & Responsibilities

Frontend Components

`app/page.tsx`

- Responsibility: Homepage and main entry point
- Features: Displays lead capture form
- Data Flow: Form → POST /api/submit

`components/lead-form.tsx`

- Responsibility: Lead capture form UI
- Fields: name, email, company, title, phone, budget, timeline, details
- Schema: `FormSchema` (Zod validation)
- Handlers: Form submission triggered on button click

`components/ui/\*`

- Responsibility: Reusable UI components (Button, Input, Textarea, Label, etc.)
- Framework: Radix UI + Tailwind CSS
- Pattern: Headless component + styling

API Routes

`app/api/submit/route.ts`

-
- POST /api/submit
- Request: FormSchema (name, email, company, title, phone, budget, timeline, details)
- Response: { success: true, message: "Lead received" }
- Purpose:
- 1. Validate form data
- 2. Create workflow instance
- 3. Return immediate confirmation
- 4. Trigger background processing

`app/api/slack/route.ts`

-
- POST /api/slack
- Purpose: Receive Slack webhook events
- Events:
- - User clicks "Approve Email" → Send email to lead
- - User clicks "Deny Email" → Discard email
- Pattern: Vercel Slack Bolt adapter handles verification

Workflow Steps (Workflow DevKit)

stepResearch(data: FormSchema)

Phase 1: Background Verification (Operational)

- Email validation and domain reputation check
- Phone number validation
- Company research via Exa.ai
- Decision-maker verification (title authority level)
- Tech stack analysis

Phase 2: Deep Research (AI Agent)

- Comprehensive company analysis
- Market positioning research
- Competitive landscape
- Technology requirements analysis

Output: Complete research report with findings

`stepQualify(data: FormSchema, research: string)`

- Takes form data + research report
- Uses `generateObject()` to extract structured qualification
- Output: `QualificationSchema` with decision and reason

`stepEmail(qualifications: QualificationSchema)`

- Uses `writeEmail()` to generate personalized response
- Creates Slack message with drafted email
- Sends to approval channel for human review

`humanFeedback()`

- Waits for Slack user interaction
- Options: Approve or Deny
- Triggers webhook for final action

5. Core Services

AI Services (`lib/services.ts`)

`deepResearch(data: FormSchema): Promise<string>`

- Model: Vercel AI Gateway (configurable)
- Purpose: Comprehensive lead research using AI agent(GPT5 mini)
- Inputs: Lead name, company, title, context
- Output: Research report (markdown)
- Uses: AI SDK agent pattern with tool use

`qualify(data: FormSchema, research: string): Promise<QualificationSchema>`

- Model: Claude or Groq (via AI Gateway)
- Purpose: Classify lead based on rules
- Output: `{ decision: 'qualified'|'unqualified', reason: string, score: number }`

`writeEmail(qualification: QualificationSchema, data: FormSchema): Promise<string>`

- Purpose: Generate personalized email response
- Tone: Professional, personalized to company research
- Output: Email body (HTML/Text)

```
# `humanFeedback(email: string, data: FormSchema): Promise<ApprovalResult>`
```

- Purpose: Send to Slack for human review
- Medium: Slack button interaction
- Returns: { approved: boolean, timestamp: string }

Research Services (`lib/background-verification.ts`)

```
# `performBackgroundVerification(data: FormSchema): Promise<VerificationResult>`
```

Comprehensive verification with 6 modules:

6. Email Validation (`validateEmail`)

- Domain format & MX records
- Domain reputation (high/medium/low)
- Temporary/disposable email detection
- Corporate vs personal domain

7. Phone Validation (`validatePhoneNumber`)

- Format validation
- Test number detection
- Sequential digit detection
- Country code validation

8. Company Research (`researchCompany`)

- Uses Exa.ai search + content extraction
- Extracts: name, industry, employees, funding, location, website
- LinkedIn company page discovery
- Error handling with fallback data

9. Decision-Maker Verification (`verifyDecisionMaker`)

- Title authority level classification:
- C-Suite (CEO, CTO, CFO, VP, President)
- VP/Director (VP, Director, Head)
- Manager (Manager, Coordinator)
- Individual Contributor
- LinkedIn profile matching
- Title-company alignment validation

10. Tech Stack Analysis (`analyzeTechStack`)

- Parses tech mentions from research
- Compatibility assessment
- Match analysis

11. Financial Health Check (`checkFinancialHealth()`)

- Funding status extraction
- Red flag detection (bankruptcy, layoffs, litigation)
- Recent news analysis

Web Search (`lib/exa.ts`)

- `exa.searchAndContents(query: string, options: ExaOptions)`
- Purpose: Web search and content extraction
- API: Exa.ai
- Returns: Search results with full page contents
- Used In: Company research, tech stack analysis, financial health

Slack Integration (`lib/slack.ts`)

Key functions:

- `sendApprovalMessage()` - Sends draft email to Slack for approval
- `handleApproval()` - Processes approve/deny button clicks
- `logToSlack()` - Posts status updates to channel

6. Data Models & Types

`FormSchema` (Lead Input)

- ```
{
 name: string; // Lead contact name
 email: string; // Email (validated)
 company: string; // Company name
 title: string; // Job title
 phone: string; // Phone number
 budget: string; // Budget range (optional)
 timeline: string; // Sales cycle timeline
 details: string; // Additional context
}
```

### `VerificationResult` (Background Check Output)

Contains 6 nested objects:

- `email`: { valid, domain, domainAge, domainReputation, mxRecords }
- `phone`: { valid, reason }
- `company`: { name, found, description, employees, funding, industry, location, website, linkedinUrl }
- `decisionMaker`: { titleValid, titleLevel, linkedinMatch, linkedinUrl }
- `techStack`: { primaryTechs[], compatibility, matchAnalysis }
- `financialHealth`: { status, redFlags[], funding, recentNews[] }
- `riskFactors`: { overall, factors[], recommendations[] }

#### **`QualificationSchema` (Lead Decision)**

- 
- {
- decision: 'qualified' | 'unqualified' | 'needs\_review';
- reason: string;
- score: number;                   // 0-100
- keyFactors: string[];   // Decision drivers
- recommendations: string[];
- }

#### **`ApprovalResult` (Email Approval)**

- 
- {
- approved: boolean;
- timestamp: string;
- reviewer: string;           // Slack user ID
- }

---

## **7. Configuration & Environment Variables**

Required `.env.local`:

- 
- # AI Services
- AI\_GATEWAY\_API\_KEY=xxx           # Vercel AI Gateway key
- GROQ\_API\_KEY=xxx                # Optional: Groq LLM
- ANTHROPIC\_API\_KEY=xxx           # Optional: Claude
- # Slack
- SLACK\_BOT\_TOKEN=xxx            # Bot token (xoxb-)
- SLACK\_SIGNING\_SECRET=xxx        # Request verification
- SLACK\_CHANNEL\_ID=xxx            # Approval channel ID
- # Research
- EXA\_API\_KEY=xxx                 # Exa.ai search API

- # Vercel
- VERCEL\_KV\_URL=xxx # Redis connection string
- VERCEL\_KV\_REST\_API\_URL=xxx
- VERCEL\_KV\_REST\_API\_TOKEN=xxx

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## 8. Workflow Execution Flow

### Timeline

1. User submits form (< 1 second)
  - Returns 200 OK immediately
  - Client shows success toast
2. Background workflow starts (Workflow DevKit)
  - All steps execute asynchronously
  - Durable execution (survives restarts)
3. Research phase (5-15 seconds)
  - Background verification runs in parallel
  - Exa.ai search and analysis
  - AI agent deep research
4. Qualification phase (2-5 seconds)
  - generateObject() structures qualification
  - Applies business rules
5. Email generation (2-3 seconds)
  - AI generates personalized email
  - Formats for Slack display
6. Human approval (Variable)
  - Waits for Slack user action
  - Can timeout after period (configurable)
7. Final delivery (< 1 second)
  - Sends email to lead (if approved)
  - Logs to Slack
  - Workflow completes

## Error Handling

- Verification fails: Uses fallback data, continues workflow
- Research timeout: Uses partial results
- Email generation fails: Posts error to Slack for manual action
- Slack approval fails: Retries with exponential backoff

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## 9. Verification Modules Deep Dive

### Email Validation Logic

- - Input: email string
  - 1. Check format (RFC 5322)
  - 2. Extract domain
  - 3. Validate MX records exist
  - 4. Check domain age (via whois estimation)
  - 5. Assess domain reputation:
    - - High: Known business domain (.com, .io)
    - - Medium: Regional domain (.co.uk)
    - - Low: Free email provider (gmail.com, yahoo.com)
  - 6. Flag temporary email services (tempmail, guerrillamail)
  - Output: VerificationResult.email

### Company Research Process

- - Input: company name from form
  - 1. Query Exa.ai: "{company} company information"
  - 2. Extract from top results:
    - - Company description
    - - Employee count
    - - Funding status
    - - Industry classification
    - - Headquarters location
    - - Official website
    - - LinkedIn company URL
  - 3. Fallback: If not found, return "not found" status
  - Output: VerificationResult.company

### Decision-Maker Classification

- - Input: job title string
  - Classification Rules:
    - - Is title in C-suite list? → C-Suite

- - Contains "VP" or "Director"? → VP/Director
- - Contains "Manager"? → Manager
- - Otherwise → Individual Contributor
- Additional checks:
  - - LinkedIn profile match (if available)
  - - Title-company alignment validation
  - - Authority level assessment
- Output: `VerificationResult.decisionMaker`

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## 10. Performance Characteristics

### Latency Breakdown (Typical)

| Component                                     | Time  | Notes                   |
|-----------------------------------------------|-------|-------------------------|
| Form submission                               | 50ms  | Client → API            |
| Workflow creation                             | 100ms | Workflow DevKit         |
| Email validation                              | 50ms  | Sync, regex + MX check  |
| Company research                              | 3-8s  | Exa.ai + parsing        |
| Decision-maker verification                   | 500ms | Sync, title matching    |
| Tech stack analysis                           | 1-2s  | Exa.ai search           |
| Financial health check                        | 2-4s  | Exa.ai search           |
| Deep research (AI Agent)                      | 5-10s | LLM + tool use          |
| Qualification ( <code>generateObject</code> ) | 2-3s  | LLM + schema extraction |
| Email generation                              | 2-3s  | LLM generation          |

- Total background time | 15-35s | Parallel where possible |

### Throughput

- Single instance can process ~10 leads/minute
- Scales horizontally via Vercel Functions
- No database bottlenecks (uses KV for state)

### Cost Drivers

- AI Gateway API calls - deepResearch, qualification, email generation
- Exa.ai searches - Company research, tech stack, financial health
- Slack API calls - Message posting, reaction handling

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## 11. Common Customization Points

### Adding New Verification Modules

1. Create new function in ``lib/background-verification.ts``
2. Add to ``VerificationResult`` interface
3. Call from ``performBackgroundVerification()``

4. Update qualification logic in `lib/services.ts`

#### Changing Qualification Logic

- Edit `lib/qualification-rules.ts`
- Modify `generateObject()` prompt for different scoring
- Adjust decision thresholds

#### Slack Integration Customization

- Channel routing: Edit `lib/slack.ts` for different approval channels based on lead criteria
- Message formatting: Adjust Slack block builder output in `stepEmail()`

#### AI Model Selection

- Change model in `lib/services.ts` by updating provider
- Use Vercel AI Gateway for multi-model routing
- Supports Claude, Groq Llama, GPT models

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## 12. Deployment Considerations

### Hosting

- Primary: Vercel (recommended for Next.js + Workflow)
- Database: Vercel KV (included with Vercel Pro)
- Build: `pnpm build` (automatically runs on Vercel)

### Environment Setup

1. Clone repository
2. Install dependencies: `pnpm install`
3. Configure `.env.local` with all required API keys
4. Deploy: `vercel deploy` or connect GitHub

### Monitoring

- Vercel Analytics: Performance metrics
- Workflow DevKit logs: Execution history
- Slack channel: Real-time approval flow logs

### Scaling Considerations

- Workflow DevKit handles concurrent leads automatically
- AI Gateway provides model failover
- KV storage is globally distributed
- Slack integration is stateless (scalable)

## 13. Testing & Validation

### Manual Testing Workflow

1. Submit test lead via form
2. Check Slack for approval message (should appear in 30s)
3. Click approve/deny button
4. Verify workflow completion in logs

### Test Data

- 
- {
- "name": "Test Lead",
- "email": "test@company.com",
- "company": "Acme Corp",
- "title": "VP Engineering",
- "phone": "+1-555-123-4567",
- "budget": "\$100k+",
- "timeline": "3-6 months",
- "details": "Interested in integration"
- }

### Common Issues & Solutions

| Issue                       | Cause                  | Solution                          |
|-----------------------------|------------------------|-----------------------------------|
| Slack message not appearing | Channel ID incorrect   | Verify SLACK_CHANNEL_ID in env    |
| Research step fails         | Exa API key invalid    | Check EXA_API_KEY in env          |
| Email not generating        | Token limit exceeded   | Check AI gateway quotas           |
| Workflow hangs              | Slack approval timeout | Manually trigger webhook endpoint |

---

## 14. File Structure Reference

- 
- lead-agent/
  - └─ app/
    - │ └─ api/
      - │ │ └─ submit/route.ts # Form submission endpoint
      - │ │ └─ slack/route.ts # Slack webhook receiver
    - │ └─ layout.tsx # Root layout
    - │ └─ page.tsx # Homepage
    - │ └─ globals.css # Global styles
  - │
  - └─ components/
    - │ └─ lead-form.tsx # Lead capture form
    - │ └─ ui/ # Radix UI components
  - │
  - └─ lib/
    - │ └─ background-verification.ts # All 6 verification modules
    - │ └─ exa.ts # Exa.ai integration
    - │ └─ services.ts # AI service calls
    - │ └─ slack.ts # Slack integration
    - │ └─ types.ts # Type definitions (FormSchema, VerificationResult, etc.)
    - │ └─ utils.ts # Helper utilities
    - │ └─ qualification-rules.ts # Lead scoring rules
  - │
  - └─ workflows/
    - │ └─ inbound/
      - │ │ └─ index.ts # Workflow definition
      - │ │ └─ steps.ts # stepResearch, stepQualify, stepEmail, etc.
  - │
  - └─ public/ # Static assets
  - └─ package.json # Dependencies
  - └─ tsconfig.json # TypeScript config
  - └─ next.config.ts # Next.js config
  - └─ tailwind.config.ts # Tailwind CSS config
  - │
  - └─ [Documentation files]
    - └─ README.md # Project overview
    - └─ START\_HERE.md # Quick start guide
    - └─ PKD.md # This file
    - └─ ARCHITECTURE\_DIAGRAM.md # Mermaid diagrams
    - └─ LEAD\_QUALIFICATION\_FRAMEWORK.md

- └─ DEBUG\_GUIDE.md # Debugging tips
- └─ CUSTOMIZATION\_GUIDE.md # How to extend

---

## 15. Key Code Patterns

### Pattern 1: Workflow Step Definition

```

•
• export const stepName = async (data: InputType) => {
• 'use step'; // Magic string for Workflow DevKit
• try {
• // Step logic here
• return result;
• } catch (error) {
• console.error('[ERROR] stepName failed:', error);
• throw error;
• }
• };

```

### Pattern 2: LLM Structured Generation

```

•
• import { generateObject } from 'ai';
• const result = await generateObject({
• model: models.default,
• schema: QualificationSchema,
• prompt: `Qualify this lead: ${JSON.stringify(data)}`
• });

```

### Pattern 3: Service with Error Handling

```

•
• async function verificationFunction(input: Input): Promise<Output> {
• try {
• const result = await externalAPI.call(input);
• return parseResult(result);
• } catch (error) {
• console.error('[ERROR]', error);
• return fallbackValue; // Graceful degradation
• }
• }

```

## Pattern 4: Slack Interaction

```
•
• const message = {
• channel: SLACK_CHANNEL_ID,
• blocks: [
• { type: "section", text: { type: "mrkdwn", text: content } },
• { type: "actions", elements: [
• { type: "button", text: "Approve", action_id: "approve" }
•]}
•]
• };
• await app.client.chat.postMessage(message);
```

---

## 16. Troubleshooting Guide

### Debugging Workflows

1. Check Vercel Function logs: `vercel logs`
2. Search for `[DEBUG]` and `[ERROR]` logs in output
3. Verify workflow instance status in Workflow DevKit dashboard

### Common Errors

- "Invalid API key" → Check environment variables
- "Slack verification failed" → Verify SLACK\_SIGNING\_SECRET
- "Workflow timeout" → Increase timeout in Workflow DevKit config
- "Search results empty" → Exa API quota exceeded

### Performance Optimization

- Cache Exa search results using Vercel KV
- Use AI Gateway for intelligent model routing
- Parallelize verification steps where possible

---

## 17. Future Enhancements

Potential improvements:

1. Save results to database (Postgres via Vercel)
2. Email delivery service integration (Resend)
3. CRM sync (HubSpot, Salesforce)
4. Real-time dashboard for lead metrics
5. Webhook integrations for external platforms
6. Custom qualification rule builder UI
7. Lead history and email tracking

## 18. Quick Reference Commands

- 
- # Development
- `pnpm install` # Install dependencies
- `pnpm dev` # Start dev server (localhost:3000)
- # Build & Deployment
- `pnpm build` # Build for production
- `pnpm start` # Start production server
- `vercel deploy` # Deploy to Vercel
- # Testing
- `curl -X POST http://localhost:3000/api/submit \`
- `-H "Content-Type: application/json" \`
- `-d '{"name":"Test","email":"test@example.com",...}'`

---

## 19. Contact & Support

- Main Documentation: See README.md
- Architecture Details: ARCHITECTURE\_DIAGRAM.md
- Customization: CUSTOMIZATION\_GUIDE.md
- Debugging: DEBUG\_GUIDE.md
- Issue Tracking: GitHub Issues

## 20. Executive Summary

Lead Agent represents a modern, production-ready solution for automating lead qualification in B2B sales environments. Built on stable, battle-tested technologies (Next.js, Vercel, Workflow DevKit), the system demonstrates several architectural best practices:

Strengths:

1. Parallel execution architecture reduces end-to-end latency
2. Graceful degradation with fallback data ensures reliability
3. Human-in-the-loop via Slack maintains organizational control
4. Multi-service integration (Exa, AI SDK, Slack) shows versatile extensibility
5. Type-safe implementation with TypeScript prevents runtime errors
6. Durable workflow execution survives infrastructure failures

Technical Innovation:

- 6-module verification system provides comprehensive lead validation
- AI-powered research layer enhances decision-making quality
- Real-time Slack integration enables distributed team collaboration
- Serverless infrastructure eliminates operational overhead

#### Cost Efficiency:

- On-demand execution reduces infrastructure costs
- Parallel processing maximizes computational efficiency
- KV-based state management eliminates database overhead

#### Customization Potential:

The modular design accommodates organization-specific requirements without major refactoring:

- Verification modules can be added or removed independently
- Qualification rules are externalized for easy adjustment
- Slack routing can be customized per lead segment
- AI model selection is configurable

#### Recommendation:

Lead Agent is suitable for organizations with 50-500 leads per month and teams already using Slack for communication. The architecture scales horizontally and supports additional integrations (CRM, email delivery, analytics) through the documented extension points.